

Improving Bioinformatic Tools for the Discovery of Plastic-Degrading Enzymes from Environmental Metagenomes

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Abstract. Plastic pollution is a major environmental challenge, and environmental metagenomes are a valuable resource for investigating the microbial potential associated with plastic transformation. This work applied and validated the M-PARTY tool to identify candidate polyethylene terephthalate (PET) degrading enzymes in publicly available metagenomes from plastic-contaminated environments. Using a reference set of experimentally verified PET-degrading enzymes obtained from the PAZy database, M-PARTY searches for homologous sequences in the metagenomic reads through read mapping with KMA. The reference enzymes were retrieved programmatically from public repositories, and the workflow was applied to two metagenomes from cultures grown on plastics. In these metagenomes, two sequences closely related to known PET-active enzymes were identified: one related to the PETase PD3 from *Bruceella anthrapi* (about 62% template coverage and 51% identity, in both datasets), and one related to the esterase PpEst from *Ectopseudomonas oleovorans* (about 28% coverage and 23% identity). Both candidates were confirmed by BLAST against the NCBI nucleotide database, where their sequences matched the genome of the organism in which each enzyme had originally been described. The number of reference enzymes provided to M-PARTY had a clear effect on the results: enlarging the reference set markedly increased the tool's ability to detect these candidates. Overall, the workflow provides a reproducible basis for detecting and prioritizing candidate plastic-degrading enzymes for subsequent validation.

Keywords: plastic · metagenomics · Hidden Markov Model · enzyme · PETase · read mapping · PAZy

1 Introduction

Plastic pollution remains a persistent environmental problem across marine, terrestrial, and engineered environments, where plastic materials can accumulate and persist for long periods of time [3]. Microplastics are especially relevant because their small size allows dispersal, persistence, and close contact with microbial communities, enabling biofilm formation. These observations have contributed to the concept of the plastisphere, the community of microorganisms

associated with plastic particles and surfaces, which may differ from surrounding assemblages in both composition and functional potential [4, 5]. The scale of the problem is considerable: global plastic production keeps growing while only a small fraction of waste is recycled. Among plastics, PET is one of the most common, and its aromatic-polyester structure makes it both persistent and a realistic target for enzymatic hydrolysis, which has motivated an active search for plastic-degrading enzymes for remediation and recycling [1, 8].

From a bioinformatics perspective, plastic-associated environments are increasingly represented in public metagenomic datasets, which make it possible to investigate whether the microbial communities living on plastics carry genes for enzymes able to act on these polymers [2]. However, finding such genes in a metagenome is not trivial, because only a small fraction of the millions of sequencing reads corresponds to plastic-active enzymes, and searching for them one by one does not scale. A more practical approach is to start from enzymes that are already known to act on a given plastic and to look for similar sequences in the metagenome in an automated way.

This is the approach implemented by M-PARTY, the tool used in this work. Based on a set of reference enzymes (here, PET-degrading enzymes obtained from the PAZy database), M-PARTY searches the metagenomic reads for homologous sequences, thereby flagging candidate enzymes present in the sample. This work focuses on poly(ethylene terephthalate) (PET), one of the most studied plastics in the context of enzymatic degradation. The first phase of the project established the strategy, the state of the art, and the proposed workflow; the present article reports the application and validation of M-PARTY on real metagenomic data, together with the results obtained and their discussion. The reference enzymes were retrieved, the tool was run on two publicly available plastic-associated metagenomes, and the resulting candidates were validated by independent sequence comparison.

1.1 Objectives

The main goal of this work was to apply, validate, and improve the M-PARTY tool for the discovery of PET-degrading enzymes in environmental metagenomes. This general goal was broken down into the following specific objectives: (i) to assemble a suitable reference set of experimentally verified PET-degrading enzymes for M-PARTY, retrieving their nucleotide coding sequences from public databases; (ii) to run M-PARTY end-to-end on real paired-end sequencing data, validating and adjusting the workflow so that it executes correctly; (iii) to apply the tool to publicly available plastic-associated metagenomes in order to identify candidate PET-degrading enzymes; and (iv) to validate the detected candidates by independent sequence comparison, assessing the reliability of the results produced by the tool.

2 Related Work and State of the Art

2.1 Plastic Biodegradation and PET-Active Enzyme Families

Plastic-associated transformation is a multi-stage process influenced by environmental exposure and microbial community structure; biofilm formation may promote physicochemical alteration and enzymatic attack. For this reason, the search for plastic-related enzymes should not be restricted to a single polymer-enzyme pair. PET degradation in particular is carried out by several related but distinct enzyme families, which is important for screening because restricting the search to one family would miss relevant candidates. The best known are the PETases, which hydrolyse PET into mono(2-hydroxyethyl) terephthalate (MHET); MHETases then complete hydrolysis into terephthalic acid and ethylene glycol [1]. Several other hydrolases act on PET, often promiscuously: cutinases, lipases, carboxylesterases, and the thermophilic cutinase-like enzymes from *Thermobifida* (such as BTA-1 and TfCut), among the most studied because of their thermostability [8, 19]. Most of these enzymes belong to the α/β -hydrolase superfamily and share a conserved catalytic triad, which explains why enzymes with different names act on the same substrate and motivates a family-aware screening approach. A metagenomic hit indicates similarity to a model or enzyme family but does not by itself demonstrate biodegradation activity under environmental conditions; even so, sequence-based screening narrows a very large search space and prioritizes candidates for later validation [2].

2.2 Computational Tools for Enzyme Detection

Several approaches can detect enzymes in complex datasets, including profile Hidden Markov Models (HMMs), domain annotation, orthology-based annotation, and metabolic pipelines. Profile HMM methods (e.g., HMMER) capture the conserved positions of an entire family rather than a single sequence, which is useful when targets are distant homologs [10]. General annotation tools like InterProScan (domains/families), eggNOG-mapper (orthology), and DRAM (metabolic potential) cover broader functional annotation [14–16], while curated databases such as PAZy and PlasticDB organize prior knowledge about plastic-related enzymes [6, 7]. For raw, unassembled metagenomes, alignment-based read mapping is also relevant: KMA maps raw reads against reference sequences and reports, per reference, the coverage, identity, and depth, making it suitable for detecting specific genes directly from reads without a prior assembly [11]. M-PARTY integrates reference construction, data retrieval, and annotation in a single framework for detecting plastic-degrading enzymes in metagenomes, and supports two complementary routes: an HMM-based search (HMMER) over assembled and translated proteins, and a read-mapping route (KMA) over raw reads at the nucleotide level (Table 1) [9]; the choice of route depends on the nature of the input data. Alternative strategies could be assembled from the tools above, for example by predicting proteins from an assembled metagenome and annotating them with InterProScan or eggNOG-mapper, or by building

custom HMM profiles and searching them with HMMER. Compared with these more general pipelines, M-PARTY has the advantage of being purpose-built for plastic-degrading enzymes and of combining reference construction and screening in a single, reproducible workflow, which is why it was adopted here.

Table 1. Selected tools for targeted enzyme detection in complex datasets.

Tool	Methodology	Input / output
M-PARTY	Targeted reference construction (Meta)genomes, proteins / re- + HMM search and read map- port tables, hits ping	
HMMER	Profile HMM search	Proteins, HMM profiles / scored matches
KMA	k-mer read alignment against	Raw reads, nucleotide refs / cov- erage, identity, depth
DRAM	Metabolic annotation of Genomes, MAGs, contigs / genomes and metagenomes	Functional and metabolic sum- maries
eggNOG- mapper	Orthology-based functional an- notation	Protein or nucleotide sequences / Ortholog assignments and func- tional annotations

2.3 Datasets and Reference Resources

Public plastic-related metagenomic datasets are spread across repositories such as SRA, BioProject, and MG-RAST, spanning marine plastispheres, enrichment cultures on plastics, wastewater plastispheres, and PET/PE microcosms. For this work, a dataset from cultures grown on plastics (BioProject PRJNA849162) was selected as the primary target, given its direct relevance to PET and its availability as paired-end runs [18]. The PAZy database was used as the reference resource for PET-active enzymes, since it compiles enzymes with experimentally verified activity together with taxonomic and accession metadata [6].

3 Materials and Methods

The workflow comprises five stages (Fig. 1): in step 1, the reference enzyme set is curated from PAZy; in step 2, the corresponding nucleotide coding sequences are retrieved; in step 3, the reference families are characterized and the screening database is prepared; in step 4, the metagenomic reads are mapped against the reference with KMA; and in step 5, the resulting candidates are validated. Each stage is described below, including the main commands used.



Fig. 1. Overview of the workflow for detecting and prioritizing candidate PET-degrading enzymes in environmental metagenomes.

3.1 Reference Enzyme Dataset

The reference set was obtained from PAZy [6]. For PET, 320 entries were retrieved, of which 312 had an associated protein sequence. For each entry, the fields used in this work were the source organism and phylum, and the sequence accessions (GenBank/EMBL CDS, RefSeq protein, UniProtKB) needed to retrieve the coding sequences. The taxonomic distribution was dominated by Actinomycetota (212), followed by Pseudomonadota (50), with smaller contributions from Bacillota, Ascomycota, and others; 18 entries had no assigned phylum (Fig. 2). The set is also functionally heterogeneous: although all entries are PET-active, they belong to different families - mostly PETases, followed by cutinases, esterases, lipases, and *Thermobifida*-type hydrolases - confirming that PET hydrolysis is not restricted to a single family and that the screening must represent several.

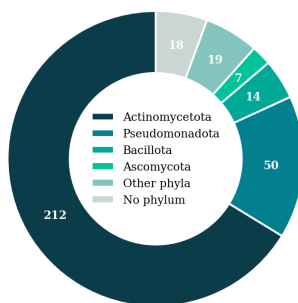


Fig. 2. Taxonomic distribution of the 320 PET reference entries from PAZy at phylum level (counts shown).

3.2 Retrieval of Nucleotide Coding Sequences

Because the read-mapping route operates at the nucleotide level, the coding sequence (CDS) of each reference enzyme was required. This was not straightforward, as most PAZy entries provide protein-level accessions. Retrieval was automated with the Biopython interface to the NCBI Entrez utilities [13], following three routes in order. First, for entries with a protein accession, the protein record was fetched and the coding region identified through the “coded_by” qualifier, which points to the genomic region and strand encoding the protein; that region was then downloaded as the CDS:

```
efetch -db protein -id <accession> -format gb
efetch -db nuccore -id <nuc_acc> -seq_start <s> -seq_stop <e> -strand <1|2> -format fasta
```

Second, for RefSeq “multispecies” protein accessions (which lack a “coded_by” field), the Identical Protein Groups (IPG) table was used to obtain a representative CDS (`efetch -db protein -id <accession> -format ipg`). Third, for entries linked only to a UniProt accession, the EMBL cross-references were used to retrieve the CDS through the same nucleotide route. Of the 320 PET entries, nucleotide coding sequences were retrieved for 257 (80%). The remaining 63 could not be retrieved automatically: 39 had no deposited sequence accession, 21 were multispecies protein records with no associated CDS in the nucleotide database, and 3 referenced inactive UniProt entries. The 257 retrieved sequences were combined into a single FASTA file used as input for the analysis.

3.3 Reference Family Characterization and Database Preparation

To organize the reference set, the 312 protein sequences were grouped by similarity with CD-HIT [12], which reduces redundancy and groups closely related enzymes into clusters; the resulting family structure is reported in Section 4.2. For the screening, the 257 nucleotide coding sequences were registered as the reference database against which the metagenomic reads would be mapped. Because the screening operates at the nucleotide level, the reference was declared as nucleic when building the database:

```
python m_party.py -w database_construction -input_seqs_db_const <references.fasta> -
input_type_db_const nucleic -hmm_db_name <db_name> -it metagenome -overwrite
```

3.4 Metagenomic Data and Annotation by Read Mapping

Two paired-end runs from cultures grown on plastics (BioProject PRJNA849162) were used as targets [18] (Table 2). They were retrieved through the data-retrieval stage of M-PARTY, which downloads the data by accession and prepares paired-end FASTQ files (`-w fetch -fetch_sra <runs> -split_files`). Screening used the read-mapping route (KMA) rather than the HMM route, because the input consisted of raw reads:

Table 2. Metagenomic runs used as screening targets (BioProject PRJNA849162).

Run	Size	Source
SRR22262629	66.7 GB	Culture grown on plastics
SRR22689345	85 GB	Culture grown on plastics

```
python m_party.py -w annotation -i <run>_1.fastq <run>_2.fastq -it metagenome -o <
output_dir> -hmm_db_name <db_name> -verbose
```

For each reference template, KMA reports the template coverage (fraction of the reference covered), the template identity, and the depth (average reads per position), written to a results table together with a reconstructed consensus sequence.

3.5 Validation of Candidates

The consensus sequence reconstructed for each hit was compared against the NCBI nucleotide database with BLAST, to confirm the identity and source organism of the detected gene and to assess whether low-coverage hits were specific. In addition, three-dimensional models of the main candidates were obtained with AlphaFold [17] and inspected for the structural fold expected for this functional family, using per-residue confidence to distinguish well-determined regions from low-confidence segments.

4 Results

4.1 Reference Set Coverage

Nucleotide coding sequences were obtained for 257 of the 320 PET reference enzymes (80%). The 63 sequences that were not retrieved were, in their majority, entries for which no nucleotide coding sequence is publicly accessible. This 257-sequence set was used as the reference for all the screening reported below.

4.2 Reference Family Structure

Clustering the 312 reference protein sequences with CD-HIT produced 146 clusters, whose size distribution was highly skewed (Table 3). The majority (98) were singletons, while a few large clusters concentrated many closely related sequences. The two largest were cluster 3, with 36 PETase-like enzymes from Actinomycetota, and cluster 42, with 20 thermophilic cutinases from *Thermobifida*. This shows that the reference set, although taxonomically dominated by Actinomycetota, is internally organized into a small number of well-populated enzyme families alongside many unique sequences.

Table 3. Largest reference enzyme clusters obtained from the 312 PET reference sequences.

Cluster	Size	Dominant character
3	36	PETase-like enzymes from Actinomycetota (<i>Micromonospora</i> , <i>Plantactinospora</i>)
42	20	Thermophilic cutinases from <i>Thermobifida</i> (BTA-1, Tfu_0882/0883, TfCut)
92	8	PETase-like enzymes from Actinomycetota
27	7	PETase-like enzymes from Actinomycetota
10	7	PETase-like enzymes from Actinomycetota

4.3 Enzyme Detection in the Metagenomes

Screening of the two metagenomes against the 257-sequence reference set produced, in each run, a few substantial hits and a larger set of low-coverage matches. Two sequences stood out clearly (Table 4, Fig. 3). The first, hereafter candidate 1, was most similar to the PETase PD3 from *Brucella anthropi* (Pseudomonadota), with coverage above 60%, identity above 50%, and depth of $\approx 46\text{-}47\times$ in both metagenomes. The second, candidate 2, was most similar to the esterase PpEst from *Ectopseudomonas oleovorans* (Pseudomonadota), detected in one run with intermediate coverage ($\approx 28\%$) and substantial depth. All remaining hits showed coverage below 5% and very low depth. These results represent a notable improvement over Phase 2, where the screening was performed against a considerably smaller reference set of only 51 enzymes. At that stage, the best hits - including *Thh_Est* (*T. halotolerans*) and *Est119* (*T. alba*) - showed coverage below 5% and identity values under 4%, making any functional inference unreliable. The expansion of the reference set to 257 sequences significantly broadened the phylogenetic coverage of the search, enabling the detection of genuinely promising candidates.

Table 4. Main candidate sequences detected by read mapping against the 257-sequence reference set, with the most similar reference enzyme for each.

Candidate	Most similar enzyme	Source organism	Metagenome	Cov. (%)	Id. (%)	Depth
Candidate 1 PD3 (PETase)		<i>Brucella anthropi</i>	SRR22262629	61.95	51.37	46.0
Candidate 1 PD3 (PETase)		<i>Brucella anthropi</i>	SRR22689345	62.32	51.65	47.1
Candidate 2 PpEst (esterase)		<i>Ectopseudomonas oleovorans</i>	SRR22689345	27.72	23.10	26.6

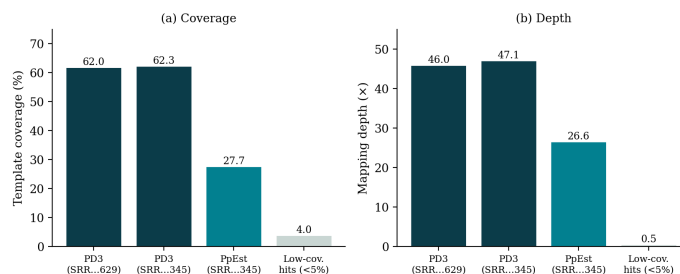


Fig. 3. Template coverage (a) and mapping depth (b) of the detected candidates versus the low-coverage background. Candidates 1 and 2 stand clearly above the background.

4.4 Validation by Sequence Comparison

The consensus sequences of the two candidates were confirmed by BLAST against the NCBI nucleotide database (Table 5). Both matched, with full identity and full query coverage, the genome of the organism in which the most similar reference enzyme had been described (*Brucella anthropi* for candidate 1 and *Ectopseudomonas oleovorans* for candidate 2), confirming a specific detection. In contrast, the low-coverage hits reconstructed only short fragments (a few tens of nucleotides) that matched unrelated genomes with no apparent relationship to plastic degradation; these were therefore interpreted as false positives. Coverage and mapping depth thus separated specific, reproducible detections from background, a separation corroborated by the sequence comparison.

Table 5. BLAST results for the consensus sequences of the two candidates against the NCBI nucleotide database.

Candidate	Best BLAST match	Identity (%)	Query cover (%)	E-value
Candidate 1	<i>Brucella anthropi</i> genome	100	100	0.0
Candidate 2	<i>Ectopseudomonas oleovorans</i> genome	100	100	0.0

4.5 Structural Models

Three-dimensional models of the two candidates were obtained with AlphaFold (Fig. 4) [17]. Both models share a common globular α/β -hydrolase domain containing the catalytic machinery, modelled with high per-residue confidence (blue), with low-confidence regions (orange) confined to a terminal extension rather than the catalytic core. The presence of this shared fold in both candidates is consistent with their common hydrolytic function on PET-related substrates; structural similarity alone does not demonstrate activity, but it reinforces their assignment to the targeted enzyme family.

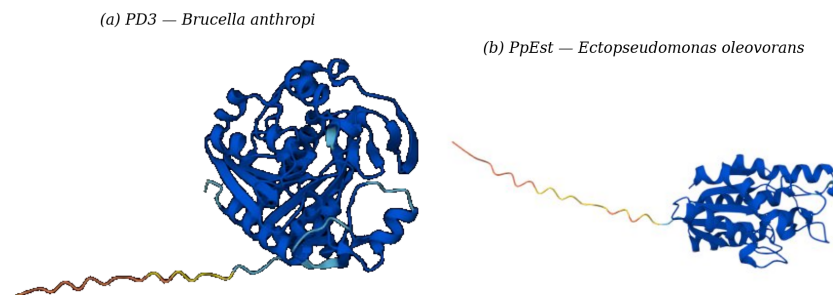


Fig. 4. AlphaFold models of the two candidates: (a) candidate 1 (most similar to PD3, from *Brucella anthropi*), and (b) candidate 2 (most similar to PpEst, from *Ectopseudomonas oleovorans*). Colours indicate per-residue confidence (blue, high; orange, low). Both share the globular α/β -hydrolase domain, with low-confidence disordered terminal segments.

5 Discussion

The results show that a targeted workflow combining a curated reference set of experimentally verified enzymes with read mapping can detect candidate plastic-degrading enzyme genes in environmental metagenomes. The two candidates were both detected with substantial coverage and depth and independently confirmed by sequence comparison (Table 4); the detection of candidate 1 in both metagenomes, with very similar metrics, indicates a reproducible rather than incidental signal. Both candidates are most similar to enzymes from Pseudomonadota, even though the reference set is dominated by Actinomycetota (whose enzymes produced only background-level hits), suggesting that the PET-active enzymes most abundant in this environment are from Pseudomonadota, and that the most represented families in a database are not necessarily the most abundant in a sample.

The number of reference enzymes provided to M-PARTY had a decisive effect on the outcome. With a small initial reference set, screening produced only low-coverage, non-specific hits and none of the substantial candidates was visible; after enlarging the reference set to 257 sequences, the two reproducible candidates emerged. This shows that the breadth of the reference set is a major determinant of the tool’s sensitivity, and that negative results obtained with incomplete reference sets must be interpreted with caution. It also justifies the effort spent on retrieving as many reference coding sequences as possible. As a complementary direction, the read-mapping route used here could be combined with an assembly-based, HMM-driven search over predicted proteins, which might capture lower-abundance candidates not detectable at the read level.

This work provides several promising indicators to guide the experimental screening phase. A metagenomic detection indicates the presence of a gene similar to a known enzyme but does not demonstrate enzymatic activity on plastics,

so the candidates should be regarded as priorities for experimental validation. The reference set, although substantially enlarged, remains incomplete because not all reference enzymes have a publicly accessible coding sequence. Finally, the analysis was based on two metagenomes from a single study, and the thresholds separating genuine from spurious hits, while supported by sequence comparison, were defined empirically.

6 Conclusions

This work applied and validated the M-PARTY tool for the discovery of PET-degrading enzyme candidates in environmental metagenomes. Starting from a curated set of experimentally verified PET-active enzymes, coding sequences were retrieved, registered as a reference database, and used to screen two plastic-associated metagenomes by read mapping. The workflow detected two specific and reproducible candidates, most similar to a PETase from *Brucella anthrophi* and to an esterase from *Ectopseudomonas oleovorans*, both confirmed by independent sequence comparison and clearly distinguished from background. Beyond these candidates, the work shows that the breadth of the reference set is a key determinant of sensitivity, and that assembling a comprehensive nucleotide reference set from a protein-centric database is itself a meaningful contribution. Future work includes extending the analysis to the complete reference set and to additional datasets, applying the workflow to other plastics, defining a more formal validation strategy, and characterizing the prioritized candidates experimentally.

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